

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 29

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Plants produce chemicals (often alkaloids) that are toxic to the herbivores that would eat them (1) • In carefully controlled amounts these chemicals have beneficial physiological effects in humans (1)	2
01.2	<i>1 mark for each point.</i> • Only a very small proportion of species have been studied for medicinal substances, so it is not known which species will be useful (1) • Conserving the habitat protects all the species in it, including those not yet studied and the species they depend on (1)	2
01.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marsupials (kangaroo / wallaby): young develop in the pouch so are easier to study than a human foetus - developmental problems (1) • Hippopotamus: hipposudoric acid skin secretion (sunscreen / antimicrobial) - burns treatment (1) • Marine sponges: proteins preventing graft rejection - organ transplants (1)	2

Question	Marking guidance	Marks
01.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres, oils, fuels, food; new food species adapted to local conditions (morama bean, yeheb, perennial maize, Kernza, bison, eland) • Biomimetics: whale flipper tubercles, shark skin, termite-mound ventilation, bird bones, gecko adhesion, burrs / Velcro, lotus self-cleaning, spider silk • Medicines and physiological research; genetic resources - crop wild relatives, Vavilov centres, gene pools, inbreeding • Ecosystem services: atmospheric composition, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination, seed dispersal, habitat provision • Evaluation: services go unnoticed because of dynamic equilibrium; unknown future value; economic vs moral / intrinsic arguments; conservation also needed for aesthetic and cultural reasons • Justified conclusion	9
	Model answer - Level 3 (9 marks)	
	Other species supply resources we use directly: wood, fibres such as cotton and wool, oils, fuels and food. Because most farmed species were domesticated long ago and are often kept where they are poorly adapted, wild species such as the drought-resistant morama bean, the yeheb tree that grows on poor soils, perennial maize that needs no ploughing, and bison that graze grassland too poor for cattle offer new, better-adapted food sources. Wild species also carry genetic resources: crop wild relatives in Vavilov centres hold genes for disease resistance and salt tolerance that inbred commercial varieties lack.	
	Biomimetics copies adaptations into technology - humpback flipper tubercles in turbine blades, shark skin in friction-reducing coatings, termite-mound convection in naturally ventilated buildings, hollow bird bones in lightweight roofs. Medicines such as taxol, quinine, AZT and diosgenin came from wild species, and physiological research on marsupials, hippo skin and sponge proteins is helping with foetal development, burns and transplant rejection. Since only a tiny fraction of species has been studied, the unknown future value argues for conserving whole habitats.	

Question	Marking guidance	Marks
	Beyond resources, other species provide services we could not replace: regulation of atmospheric gases, evapotranspiration driving rainfall, the carbon, nitrogen and phosphorus cycles run by microbes, soil formation by decomposers, and pollination, seed dispersal and habitat provision. These are unnoticed precisely because they are in dynamic equilibrium, so their economic value is underestimated. On balance the resource and service arguments alone justify conservation, but they are strongest when combined with the intrinsic and aesthetic case, because many species with no known use still sustain the systems we depend on.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important because we get lots of things from other species. We get wood, cotton and food from plants and animals. Some medicines come from plants, like aspirin from willow trees. Animals can help design things, like Velcro was copied from seeds. Other species also do things for us like bees pollinating crops. If we lose species we might lose things that could be useful in the future. So we should conserve biodiversity.	
	What makes the difference	
	Both answers cover resources, medicines, biomimetics and services, but Level 3 develops each with specific, accurate detail (which species, what property, why it matters).	
	The question asks 'to what extent'. Level 3 weighs resource arguments against intrinsic value and reaches a reasoned conclusion; Level 1 asserts 'so we should conserve'.	
	Level 3 explains <i>why</i> services are undervalued (dynamic equilibrium) - an explanation, not just a list.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	
02.1	<i>Correct answer only.</i> B (1)	1
02.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - Harvesting no faster than the MSY - resource never depleted (1) - Mixed species plantations - more food choices, relationships and niches (1) - Indigenous species - support indigenous insects, birds (1) - Mixed age structure - varied light, temperature, wind so more species; continuous cover and yield (1) - Selective logging - small clearings, less disturbance, long-term timber benefit (1) - Retention of dead wood, buffer zones, corridors (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
03.1	<i>1 mark.</i> Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis) (1)	1

Question	Marking guidance	Marks
03.2	<i>1 mark for each point.</i> • Zebu tolerate heat but have a low milk yield; Ayrshire have a high milk yield but are not heat tolerant (1) • Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield (1)	2
03.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Replacement of valuable animals that die with genetically identical individuals (1) • Replacement of herds culled during a disease outbreak (1) • Production of large numbers of individuals with desirable characteristics (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
04.1	<i>1 mark for each point.</i> • Hard release: individuals released with no post-release support - suitable for species whose behaviour is instinctive and need no learned skills (insects, fish, amphibians, reptiles) (1) • Soft release: gradual release into larger areas with food provided while they learn to find food, avoid predators - needed for mammals and birds (1) • Problems faced: finding / recognising food and poisonous foods, hunting skills (captive-born predators never learned), avoiding predators, acceptance into wild social groups (1)	3
04.2	<i>Any 5 from the points below - 1 mark each, maximum 5.</i> • Large enough suitable habitat for a viable population (1) • Reliable food supply (1) • Low predation risk / original threat removed (1) • Suitable breeding sites (1) • Water (1) • Support of the local human population (1) • Official / legal protection of habitat and species (1)	5
04.3	<i>1 mark for each point.</i> • Tracking (radio / GPS tags, rings) shows survival, dispersal and breeding of released animals so the success of the project can be assessed (1) • Identifies causes of failure so future releases can be improved / extra support given (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 36-37; AQA spec 3.1.2.3.3.	
05.1	<i>1 mark for each point.</i> • Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud (1) • Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores); salt excretion (1) • Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration (1)	3

Question	Marking guidance	Marks
05.2	<i>1 mark for each point.</i> • Coastal erosion / storm protection: roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact) (1) • Fisheries: nursery grounds where dense roots protect young fish that later live on reefs / open sea (1) • Protection of coral reefs: slow river water so suspended solids settle and nutrients are absorbed before reaching reefs (1) • Resources: timber, fuel; medicines (antimicrobials against MRSA, cyanobacterium for lung cancer); high biodiversity (crabs, shrimps, sponges, birds) (1)	4
05.3	<i>1 mark for each point.</i> • Clearance for aquaculture (shrimp ponds), urban development, ports, tourism; pollution; sea level rise / climate change; destruction of coral reefs that shelter the coast (1) • Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015) (1) • Replanting and natural regeneration; control of damaging activities; Ramsar sites (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.	
06.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Image databases: identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features (2) • eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance; contamination / DNA transport by water (2) • Geolocator tags: tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate; animal must be recaptured to download data (2)	6
06.2	<i>1 mark for each point.</i> • Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface) (1) • Microwaves pass through cloud so monitoring is not interrupted by weather (1)	2
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Droppings: diet, gender, territory size (1) • Owl pellets: diet (1) • Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
07.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Designation of protected areas (SSSIs, MCZs) (1) • Protection of wild birds and their nests (except some pests / game birds) (1) • Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts (1) • Uprooting wild plants generally illegal (1) • Bats must not be disturbed; roof timber treatments must be non-toxic to bats (1)	4
07.2	<i>1 mark for each point.</i> • Appendix I: species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding (1) • Examples: great apes, big cats, rhinos, blue whale (1) • Appendix II: species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected (1) • Examples: Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap (1)	4
07.3	<i>1 mark for each point.</i> • Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market (1) • Removing the profit removes the incentive to collect (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.	
08.1	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Hydrological cycle (1) • Carbon cycle (1) • Nitrogen cycle (1) • Atmospheric circulation (1) • Thermohaline circulation (1) • Photosynthesis (1)	2
08.2	<i>1 mark for each point.</i> • Enzymes reduce the activation energy of reactions (1) • So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures (1)	2
08.3	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> • Haber process (nitrogen fixation) (1) • Incineration of wastes (decomposition) (1) • Many manufacturing processes (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2.	
09.1	<i>1 mark for each point.</i> • Images / maps of large areas; land-use change, deforestation, flooding (1) • Vegetation density and surface temperature (1) • Wind velocity, sea-surface altitude, wave height, oil pollution (1)	3
09.2	<i>1 mark.</i> • Microwaves pass through clouds (unlike visible light) (1)	1

Question	Marking guidance	Marks
09.3	<i>1 mark for each point.</i> • Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected (1) • No need for researchers to visit inaccessible areas, reducing cost and disturbance; data are standardised across the whole area (1)	2
09.4	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night (1) • Each tiger has a unique pattern of facial / body stripes (1) • Images are compared with the database so individuals are recognised rather than counted twice (1) • The number of different individuals recorded gives a minimum population; repeated captures give territory size, movements, lifespans and social groupings (1) • Limitation: only works for species with unique visible features (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6.	
10.1	<i>1 mark.</i> • Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (1)	1
10.2	<i>1 mark for each point.</i> • Negative feedback resists change / returns the system towards its previous state (1) • eg increased temperature increases cloud cover and albedo, or higher CO₂ increases photosynthesis (1) • Positive feedback increases the change / makes it self-sustaining (1) • eg melting permafrost releasing methane, declining albedo as ice melts, forest and peat fires (1)	4
10.3	<i>1 mark.</i> • Many species with a complex food web so the loss of one has a small effect; agroecosystems have few species and lack detritivores, pollinators and pest predators that maintain stability (1)	1

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Principles of natural systems: dynamic equilibria and negative feedback; diverse systems resistant to change; low energy-density renewable energy (solar-driven hydrological, carbon and nitrogen cycles); low-temperature processes using enzymes; material cycles with wastes as raw materials; small number of abundant elements; biodegradable non-toxic wastes • Human systems: non-renewable energy, high temperatures (Haber process, incineration, manufacturing), linear resource use, dispersal of minerals and mixtures that are hard to separate, persistent toxins (PCBs, organochlorines, toxic metals in electronics) • Applications: renewable energy and heat pumps, bioleaching, recycling and design for re-use, industrial symbiosis, organic and integrated agriculture, retaining hedgerows and pollinators, circular economy • Human responses that override feedback (fertilisers, irrigation, pesticides) and positive feedback triggered by human activity (permafrost, methane hydrates, fires); tipping points • Barriers: population size and affluence, attitudes of urban populations to distant resources, economics, need for international cooperation, existing infrastructure • Justified conclusion on how far natural principles can be applied	9
	Model answer - Level 3 (8 marks)	
	Natural systems have remained sustainable because they run on low energy-density renewable energy, mainly sunlight driving the hydrological, carbon and nitrogen cycles; their reactions occur at low temperatures because enzymes lower activation energies; materials cycle so the waste of one process is the raw material of the next; only a few abundant elements are used; and wastes are biodegradable and non-toxic. Negative feedback keeps them in dynamic equilibrium and high diversity makes them resistant to change.	
	Human systems contrast on every point: non-renewable fossil fuels, high-temperature processes such as the Haber process and incineration, linear use that disperses minerals and mixes materials that cannot be separated, and persistent toxins like PCBs and the metals in electronics. We also override negative feedback - when degraded soil reduces yields we add fertiliser, irrigation and pesticides - and trigger positive feedback such as permafrost melting.	

Question	Marking guidance	Marks
	Copying natural principles is already possible in many areas: renewable energy and heat pumps, bioleaching at low temperature, recycling and design for disassembly, industrial symbiosis, legumes and organic matter instead of manufactured nitrate, and retaining hedgerows for pest predators - the basis of the circular economy. The limits are population size and affluence, the attitude of urban consumers who never see depleted resources, cost, and the need for international cooperation. So the principles can make individual activities far more sustainable, but only a large-scale, cooperative shift in lifestyle would make the whole human system sustainable.	
	Weaker answer - Level 2 (5 marks)	
	Natural systems use renewable energy from the Sun and recycle materials, so nothing runs out and there is no waste. Humans use fossil fuels which will run out and produce waste that is toxic. Natural processes happen at low temperatures but the Haber process needs high temperatures. Humans could copy natural systems by using renewable energy like solar and wind, recycling more and using less fertiliser. This would make human activities more sustainable. However it would be expensive and people would have to change their lifestyle.	
	What makes the difference	
	• Level 2 states the contrast correctly but does not explain the mechanisms (enzymes, negative feedback, diversity, dispersal of minerals) that Level 3 uses.	
	• Level 3 applies the principles to specific technologies (bioleaching, heat pumps, industrial symbiosis, legumes); Level 2 stays at 'recycle more'.	
	• 'Evaluate the extent' - Level 3 distinguishes individual activities from the whole system in its conclusion; Level 2's evaluation is a single sentence about cost.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-374; AQA spec 3.6.1-3.6.3.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Definition of sustainability (Brundtland); population x per capita impact; dynamic equilibria, feedback, tipping points, diversity • Principles of natural systems: renewable low-density energy, low temperatures, material cycles, non-toxic wastes; contrast with linear human systems • Circular economy principles: cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies • Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids) • Measuring progress: ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index • Barriers: economics, competition, consumer behaviour, need for international agreement, existing infrastructure • Judgement on the extent achievable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-389; AQA spec 3.6.	25

Question	Marking guidance	Marks
11.2	Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Species selection: market, range of tolerance, controllability • Abiotic control: temperature, light, water, soil fertility / nutrients, aeration, salinity, pH, CO₂, topography, relief, wind - methods and examples • Biotic control: pest groups, cultural control, biological control (successes and failures), sterile males, pheromones, genetic resistance, pesticides (properties), antibiotics, pollinators, soil biota • Manipulation of food species: population density, monocultures, selective breeding, asexual reproduction, cross-breeding, AI / embryo transfer, GM; energetics - intensive vs extensive, energy subsidies, ratios, FCR, food chain losses • Impacts: habitat loss / change, reduced biodiversity, introduced species, pesticide and nutrient pollution, GHGs, hydrological changes, soil erosion; positive: plagioclimax habitats • Social, economic and political influences shaping methods • Sustainable strategies and evaluation; conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-337; AQA spec 3.5.1.	25